

PROJECT SYNOPSIS

Multimodal Presentation Attack and Injection Detection in Electronic Know Your Customer (eKYC) Architectures

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Project Type	Academic Proof-of-Concept / Research Project

1. Introduction

Remote eKYC improves the speed and convenience of identity verification, but it also creates opportunities for replay attacks, face swaps, deepfakes, voice cloning and camera-injection attacks. A single liveness mechanism may not be sufficient against modern manipulated media.

2. Problem Statement

Existing remote identity verification commonly relies on face matching and liveness checks. High-quality synthetic media and logical injection attacks can challenge these controls. This project therefore investigates a layered, multimodal approach using video, facial, behavioural, audio and audio-visual signals.

3. Aim

To design and implement a lightweight multimodal eKYC spoof-detection proof-of-concept that analyses verification media and produces an interpretable risk assessment for potential presentation, deepfake and audio-visual spoofing attacks.

4. Objectives

- Analyse uploaded eKYC videos using Python and OpenCV.
- Perform face presence and basic behavioural/liveness-related analysis.
- Use labelled secondary video data for deepfake testing.
- Add genuine/spoofed speech analysis.
- Investigate audio-visual/lip-sync consistency.
- Combine available signals through a risk-scoring engine.
- Present results using a Streamlit dashboard.
- Evaluate applicable modules using suitable metrics such as APCER, BPCER, EER and confusion-matrix measures.

5. Proposed Methodology

The system follows a layered architecture. Uploaded media is preprocessed, followed by face and liveness-related checks, video/deepfake analysis and, where audio is available, speech spoofing and audio-visual consistency analysis. The observations are combined by a risk engine producing ACCEPT, REVIEW or REJECT as a prototype assessment.

6. System Architecture

Streamlit UI → Video/Audio Input → OpenCV Processing → Face Analysis → Liveness/Behaviour → Deepfake/Video Analysis → Audio Analysis → Lip-Sync / Audio-Visual Consistency → Risk Engine → ACCEPT / REVIEW / REJECT.

7. Secondary Data and Datasets

The practical prototype uses manageable secondary datasets. SDFVD is the main video dataset, containing 106 labelled videos (53 Fake and 53 Real) with a reported total size of approximately 183 MB. A small speech deepfake demonstration dataset is used for the audio component. HAV-DF remains a relevant multimodal research reference but is not required for the initial prototype. eKYC-DF is retained as the principal eKYC-focused research reference; its current repository README lists Usage as “Coming soon!”.

8. Data Organisation

The project stores genuine videos, deepfake videos and audio samples in separate local directories. Additional genuine webcam recordings may be collected for controlled testing. Dataset labels and metadata will be retained to support reproducible evaluation.

9. Tools and Technologies

Python; Streamlit; OpenCV; NumPy; Pandas; and selected pre-trained or established detection models introduced only when required. Development and testing will be performed locally on a single Windows laptop.

10. Testing Strategy

Testing will use labelled genuine and manipulated samples, together with controlled local recordings. Tests will progressively cover video loading, face analysis, liveness-related behaviour, deepfake evidence, speech spoofing and audio-visual consistency. Known dataset labels will be used as ground truth where available.

11. Risk Assessment

A simple weighted risk engine will combine module outputs. Evidence from multiple independent modalities can increase risk, while clean observations can reduce it. REVIEW is included for uncertain cases. The weights and thresholds will be documented as prototype design choices, not as validated production thresholds.

12. Evaluation Metrics

Where ground-truth labels are available, accuracy, precision, recall, F1-score and confusion matrices may be reported. APCER and BPCER are relevant to presentation-attack evaluation, while EER is relevant to speech anti-spoofing. Metrics will only be reported for modules that are actually implemented and tested.

13. Expected Outcome

A working Streamlit demonstration that accepts eKYC media, performs multiple analysis stages, displays individual findings and produces an overall risk assessment. The project demonstrates how layered multimodal analysis can strengthen remote identity verification.

14. Scope and Limitations

This is an academic proof-of-concept, not a production KYC platform. It will not attack real financial or KYC systems. Limited hardware, dataset access and the use of manageable secondary datasets restrict experimental scale. The project focuses on architecture, implementation and evaluation methodology rather than production-level performance claims.

15. Future Work

Future work can integrate stronger deepfake classifiers, dedicated face anti-spoofing, robust voice anti-spoofing, document verification, camera-injection detection, improved lip-sync analysis, larger benchmark datasets and calibrated risk thresholds. eKYC-DF can be evaluated more extensively if full dataset access becomes available.

16. Conclusion

A multimodal and layered architecture provides broader coverage than relying on a single liveness mechanism. Combining facial, behavioural, video/deepfake, voice and audio-visual signals within an interpretable risk engine provides a practical academic approach for studying presentation and injection attacks in eKYC.

17. Project Workflow

Stage	Activity	Output
1	Upload eKYC media	Input video/audio
2	OpenCV preprocessing	Frames, FPS, duration
3	Face analysis	Face observations
4	Liveness / behaviour	Liveness-related evidence
5	Video/deepfake analysis	Real/fake evidence
6	Audio analysis	Genuine/spoof evidence
7	Lip-sync / A-V consistency	Consistency evidence
8	Risk engine	Risk score + decision
9	Dashboard and logs	Interpretable report

18. Key References

- Felouat, H., Nguyen, H. H., Le, T.-N., Yamagishi, J., and Echizen, I., “eKYC-DF: A Large-Scale Deepfake Dataset for Developing and Evaluating eKYC Systems,” IEEE Access, 2024, vol. 12, pp. 30876–30892.
- ISO/IEC 30107-3:2023, Biometric presentation attack detection — Part 3: Testing and reporting.
- AASIST / AASIST3 literature on audio anti-spoofing and speech deepfake detection.
- SyncNet and related audio-visual synchronization literature.
- Reserve Bank of India, Master Direction – Know Your Customer (KYC) Direction, including Video-based Customer Identification Process (V-CIP).
- Hemgg, SDFVD-video-dataset, Hugging Face.
- Behavioral Signals, deepfake-detection-demo, Hugging Face.